RAP managed 4 Brownfield apps

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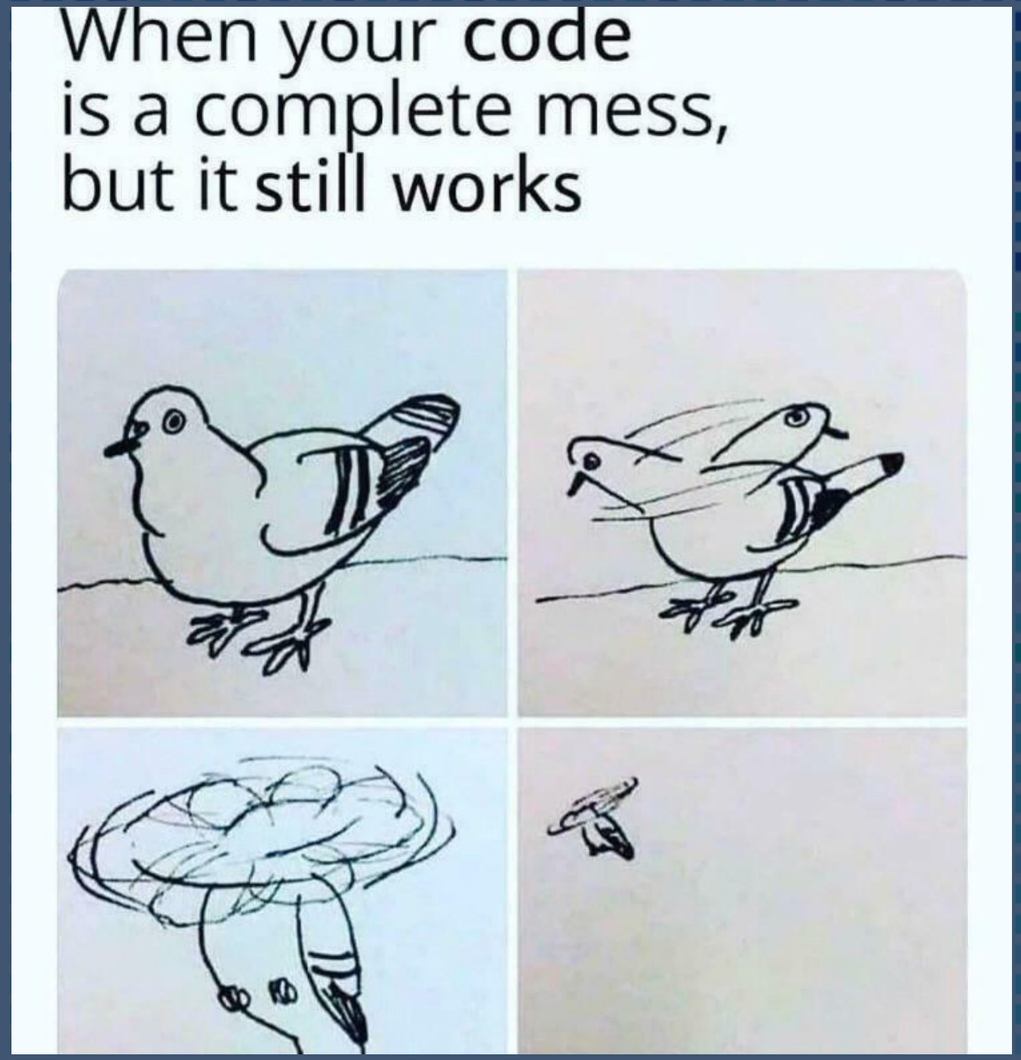
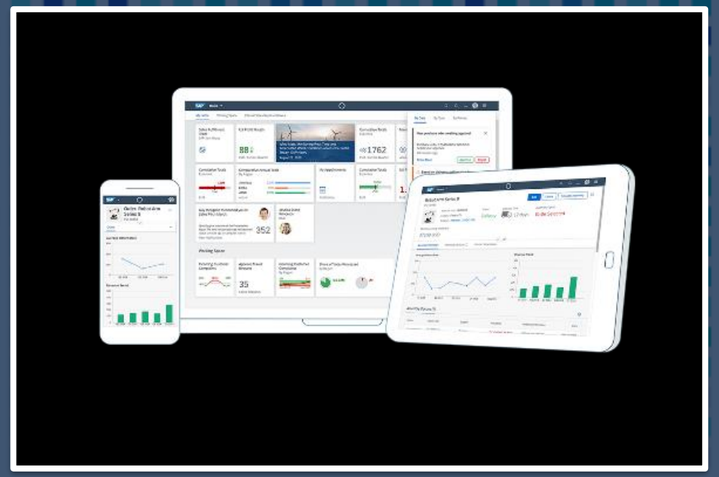
No thanks!
We are too busy


When your code
is a complete mess,
but it still works

Solution in detail AS-IS

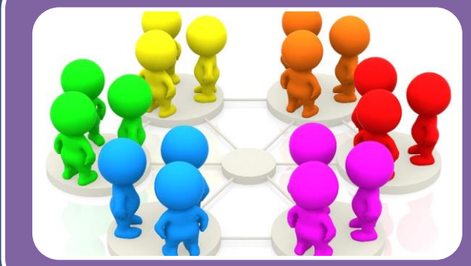


SEGW

Custom table

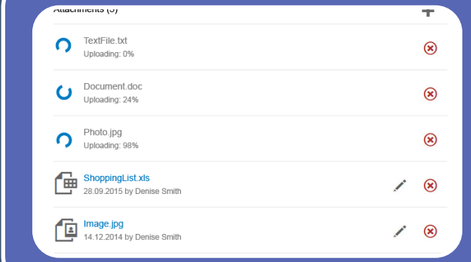
HCM
(S/4 2020)

Use-case: Onboarding app



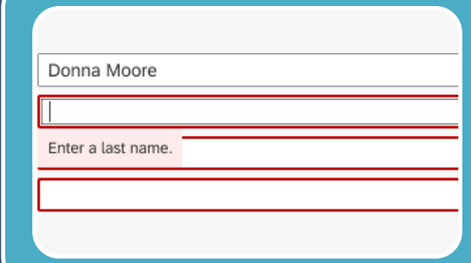
Multiple user-groups (Adapted/expanded regularly)

- End-user: Manager, manager-assistant, team-lead, …
- Onboarding groups: Internals, externals, office-attendants, operatives, …



Document upload

- Pictures
- General document (safety doc., GDPR, ...)



Draft possibility

- Start with onboarding & continue later
- Avoid people being onboarded multiple times

1 Personal Data — 2 Work Data — 3 Access Data — 4 Attachments — 5 Review & Submit

1. Personal Data

Every external person working @ must have a valid and approved shopping cart or purchase order.
It is mandatory to fill in **one of the fields: Shopping cart or Purchase order**.

Last name:* Developer Day

First name:* Flexso

Date of birth:* 15/06/2000

Language:*

Form-of-address:* Unknown Male Female Non-Binary Undeclared

PurchaseOrder / Item:* 0

ShoppingCart / Item:* 0

Company:*

Save draft | Delete | Previous | Next Step

Goal

Future proof

- Prepare for new user groups
- Easier to expand code base

Improve code quality

- Move business rules
- Remove spaghetti code
- Optimize Search helps

UI Update

- UI5 v1.41 → v1.117
- Old UI components (upload)
- Custom control (wizard)

Approach

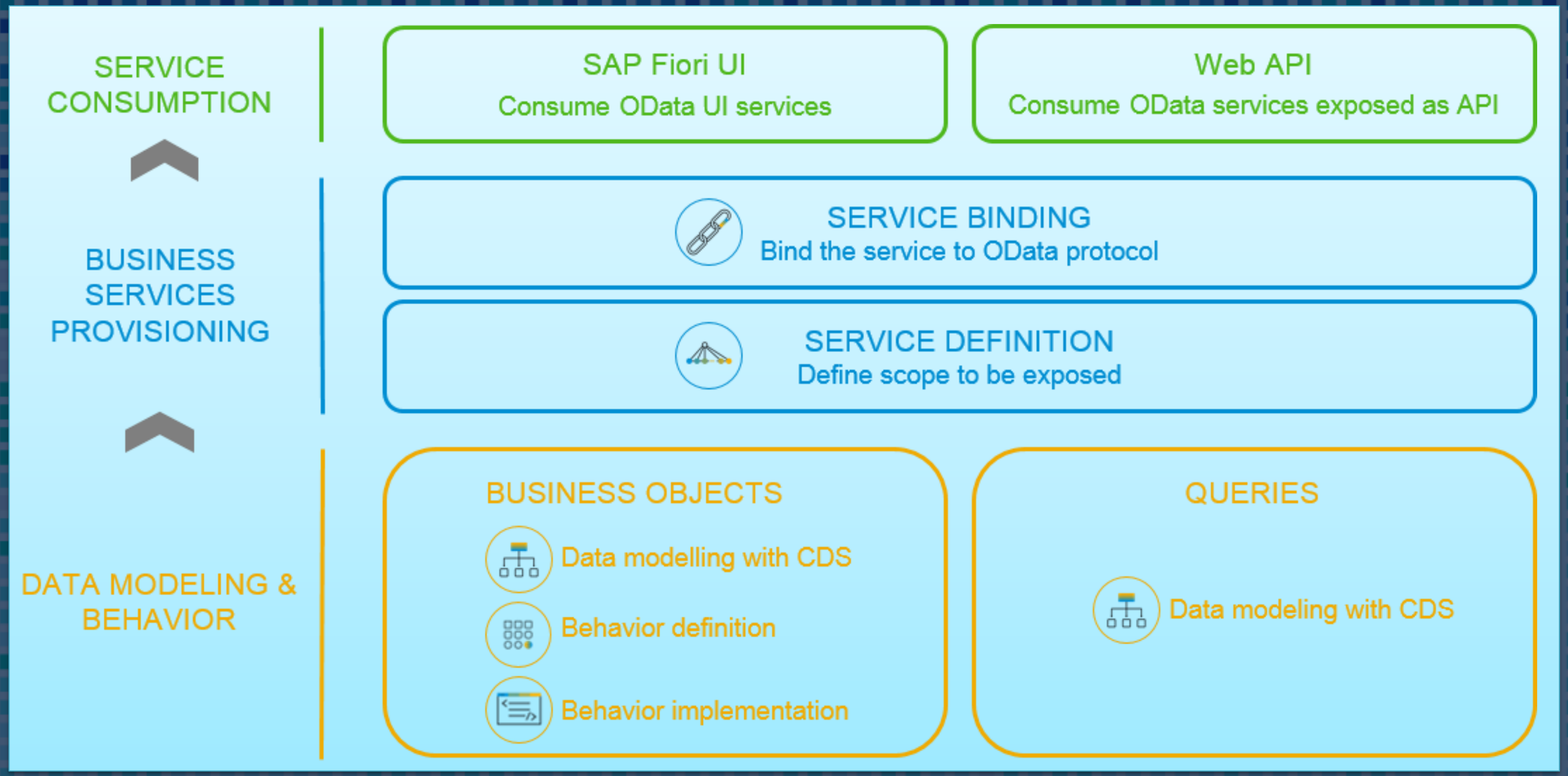
Keep

- Back-end (engine)
  - Creating HR Personnel file
  - Approval workflow

Refactor

- Front-end UI
- Data validation
- Search-helps & Drop-downs
- Draft handling
- Image/docu upload

RAP


SERVICE CONSUMPTION
SAP Fiori UI
Consume OData UI services
Web API
Consume OData services exposed as API
BUSINESS SERVICES PROVISIONING
SERVICE BINDING
Bind the service to OData protocol
SERVICE DEFINITION
Define scope to be exposed
DATA MODELING & BEHAVIOR
BUSINESS OBJECTS
Data modelling with CDS
Behavior definition
Behavior implementation
QUERIES
Data modeling with CDS

RAP

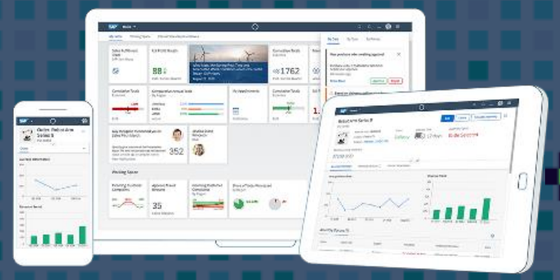
	Managed	Unmanaged
Draft handling		
Actions/Determ./Valid.		
Auto CRUD		
SAP Recommendation	Greenfield	Brownfield

Solution

Solution

SAP HCM

RAP Managed scenario with draft

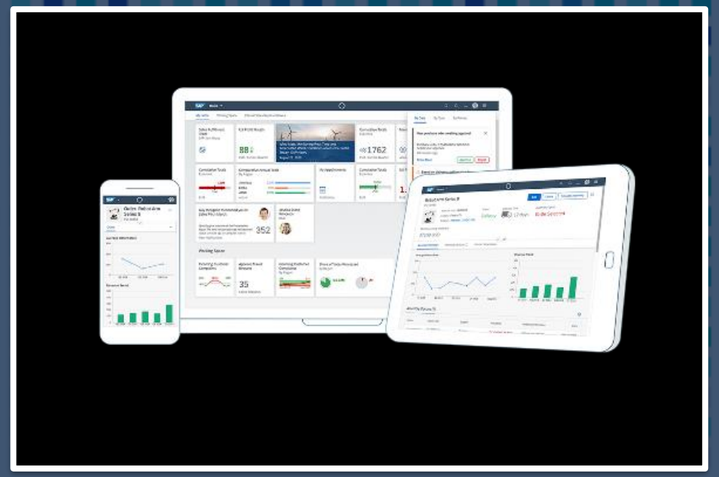


Custom table

RAP Action (submit)

SAP HCM

Solution in detail AS-IS



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Solution in detail AS-IS → TO-BE

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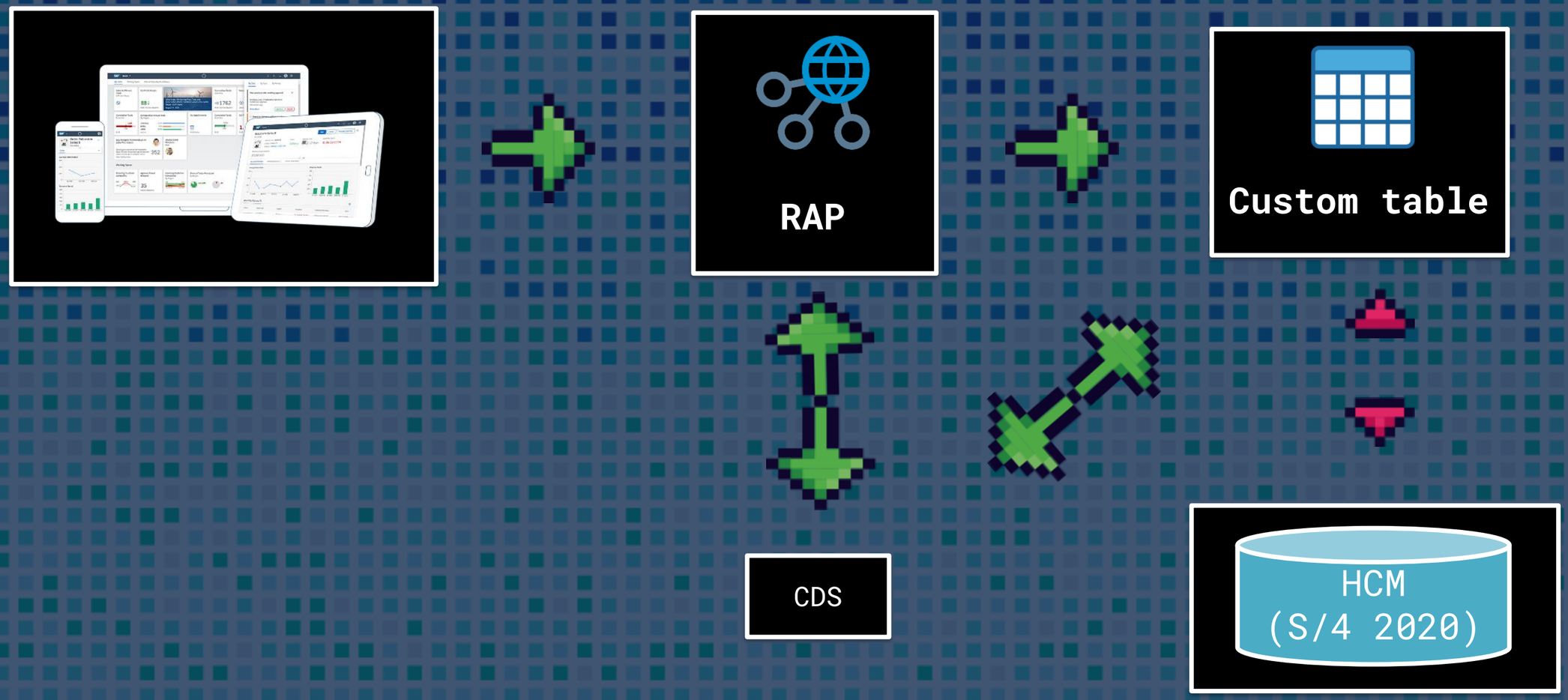


Main benefits: RAP functionalities

- Auto CrUD
  (managed)

```
managed; // in
with draft;
```

- Draft handling

- Validations & determinations
  (Business rules)

```
validation phoneNumberValidation on save { field PhoneNumber; }

determination generateInitialValues on modify { field Id; }
```

- Actions
  (vs Function imports)

```
action submitCheckIn result [1] $self;
action reGenerateWindowsUser result [1] $self;
action getManager result [1] $self;
```

Main benefits: RAP messages

- Trigger = validation
- Perform validations *(& collect messages)*

```
TRY.
    zcl_hr_chinv2_validator=>get_instance( )->validate_save( is_check_in = <ls_checkin>-%data
                                                             it_attachments = CORRESPONDING #( lt_attachments )
                                                             it_SiteAccess = CORRESPONDING #( lt_SiteAccess )
                                                            ).
  CATCH zcx_hr_chinv2 INTO DATA(lzcx_chinv2_error).
```

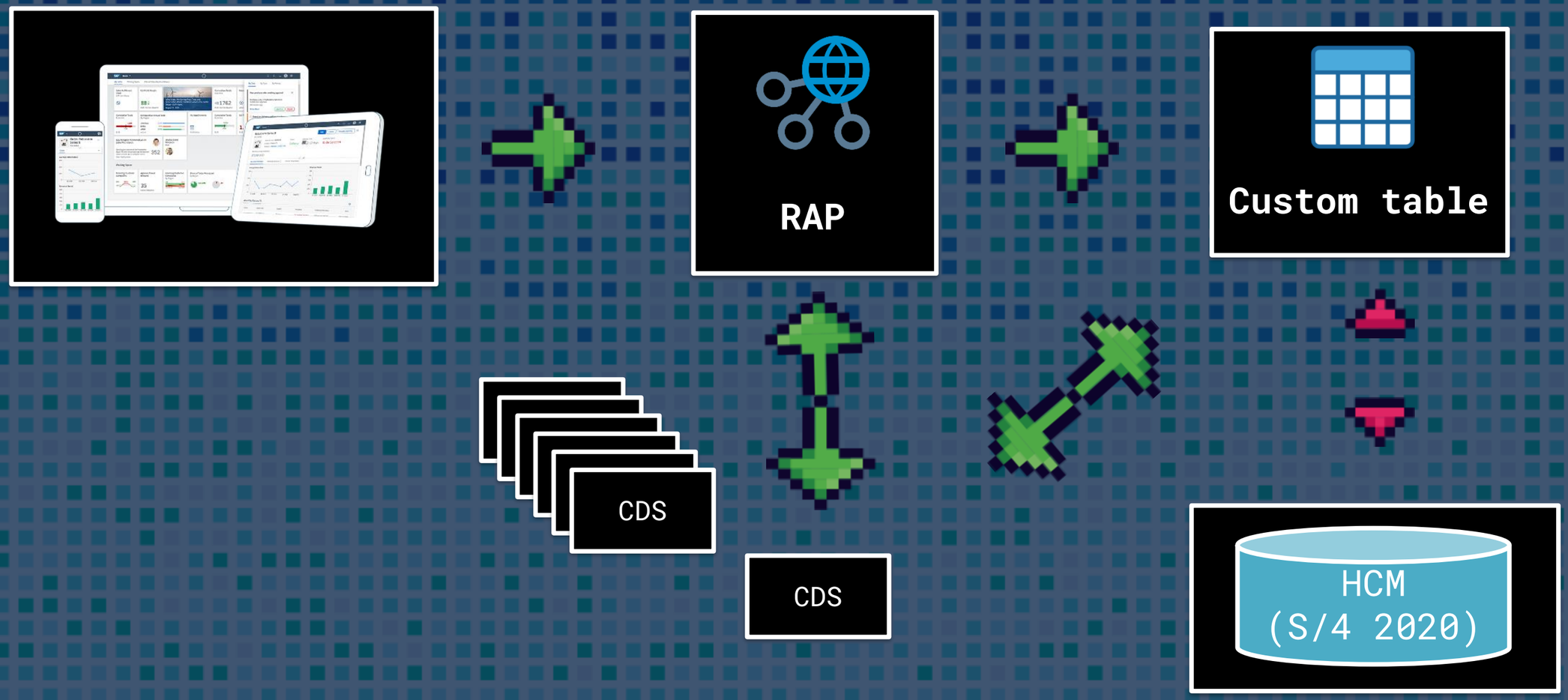
- “Reported” struc. & field ident.

```
APPEND INITIAL LINE TO reported-zhr_c_chin_checkin ASSIGNING FIELD-SYMBOL(<ls_rep_checkin>).
IF <ls_message>-field IS NOT INITIAL.
  lv_fieldname = to_upper( <ls_message>-field ).
  ASSIGN COMPONENT lv_fieldname OF STRUCTURE <ls_rep_checkin>-%element TO FIELD-SYMBOL(<lv_field>).
  <lv_field> = if_abap_behv=>mk-on.
ENDIF.
```

- Attach message

```
<ls_rep_checkin>-%tky =  <ls_checkin>-%tky.
<ls_rep_checkin>-%msg = new_message( id       = <ls_message>-id
                                     number   = <ls_message>-number
                                     v1       = <ls_message>-message_v1
                                     v2       = <ls_message>-message_v2
                                     v3       = <ls_message>-message_v3
                                     v4       = <ls_message>-message_v4
                                     severity = CONV #( <ls_message>-type ) ).
```

Solution in detail AS-IS → TO-BE



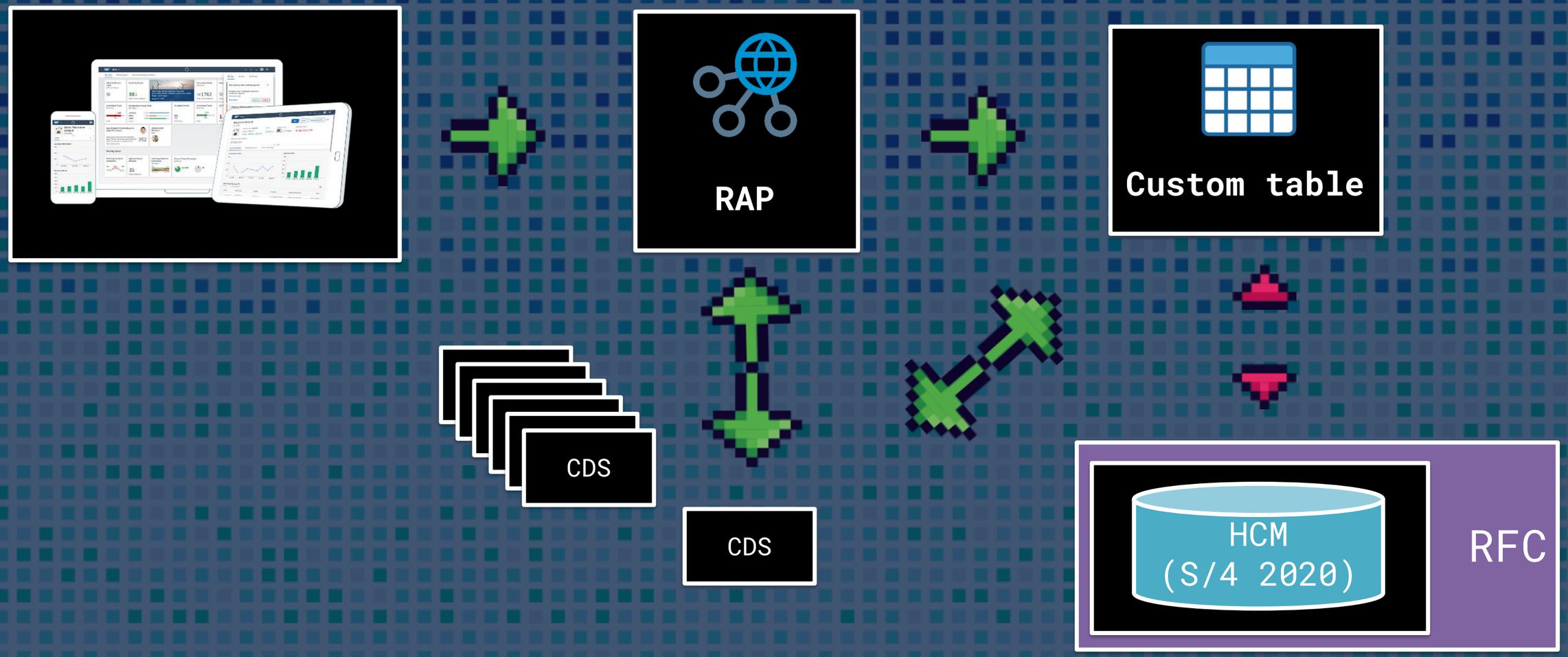
Main benefits: CDS

- Reading data
  (Easy data modelling)
- *“18!”* Valuehelps/drop-downs
- Filter/sort/… out of the box

[ED1] ZZ_HR_CHECKIN_V2 [ED1] ZZ_HR_CHECKIN_V2

```
@EndUserText.label: 'HR Check In V2'
define service ZZ_HR_CHECKIN_V2 {
  expose ZHR_C_CHIN_CHECKIN as CheckIn;
  expose ZHR_C_CHIN_CHECKIN_TILECOUNT as CheckInCount;
  expose ZHR_C_CHIN_CHECKIN_ATTACHMENT as CheckInAttachment;
  expose ZHR_C_CHIN_CHECKIN_SITE_ACCESS as CheckInSiteAccess;
  expose ZHR_C_CHIN_APPFIELDCONFIG as AppFieldConfig;
  expose ZHR_C_CHIN_CANDIDATES_VH as Employees_vh;
  expose ZHR_C_CHIN_SITEON1STDAY_VH as SiteOn1stday_vh;
  expose ZHR_C_CHIN_50HZ_USER_LOCATIONS as UserLocation50HZ_vh;
  expose ZHR_C_CHIN_SITES_VH as Sites_vh;
  expose ZHR_C_CHIN_FORMOFADR_VH as FormOfAddress_vh;
  expose ZHR_C_CHIN_COMPANIES_VH as Companies_vh;
  expose ZHR_C_CHIN_WORKLOCATION_VH as WorkLocations_vh;
  expose ZHR_C_CHIN_LANGUAGES_VH as Languages_vh;
  expose ZHR_C_CHIN_MANAGERS_VH as Managers_vh;
  expose ZHR_C_CHIN_TEAMASSISTANT as TeamAssistant_vh;
  expose ZHR_C_CHIN_VENDORS_VH as Vendors_vh;
  expose ZHR_C_CHIN_HU_VH as HU_vh;
  expose ZHR_C_CHIN_PO_SC_ITEMS_VH as POItemsSc_vh;
  expose ZHR_C_CHIN_SC_ITEMS_VH as SCItems_vh;
  expose ZHR_C_CHIN_TYPE_OF_50H_USER_VH as TypeOfUser_vh;
  expose ZHR_C_CHIN_TYPE_OF_IDCARD_VH as IdCard_vh;
  expose ZHR_C_CHIN_TYPE_OF_USRIDCRD_VH as UserIdCard_vh;
  expose ZHR_C_CHIN_MAILBOX_VH as MailBox_vh;
}
```

Solution in detail AS-IS → TO-BE



Using RFC Function Module

- Encapsulate code
- LUW
  (COMMIT & ROLLBACK)
- Complete picture

```
CALL FUNCTION 'ZHR_CHINV2_RAP'
  DESTINATION 'NONE'
  EXPORTING
    is_checkin_v1          = ls_checkin_v1
  IMPORTING
    et_non_disrupting_msg = et_bapirret2.
```

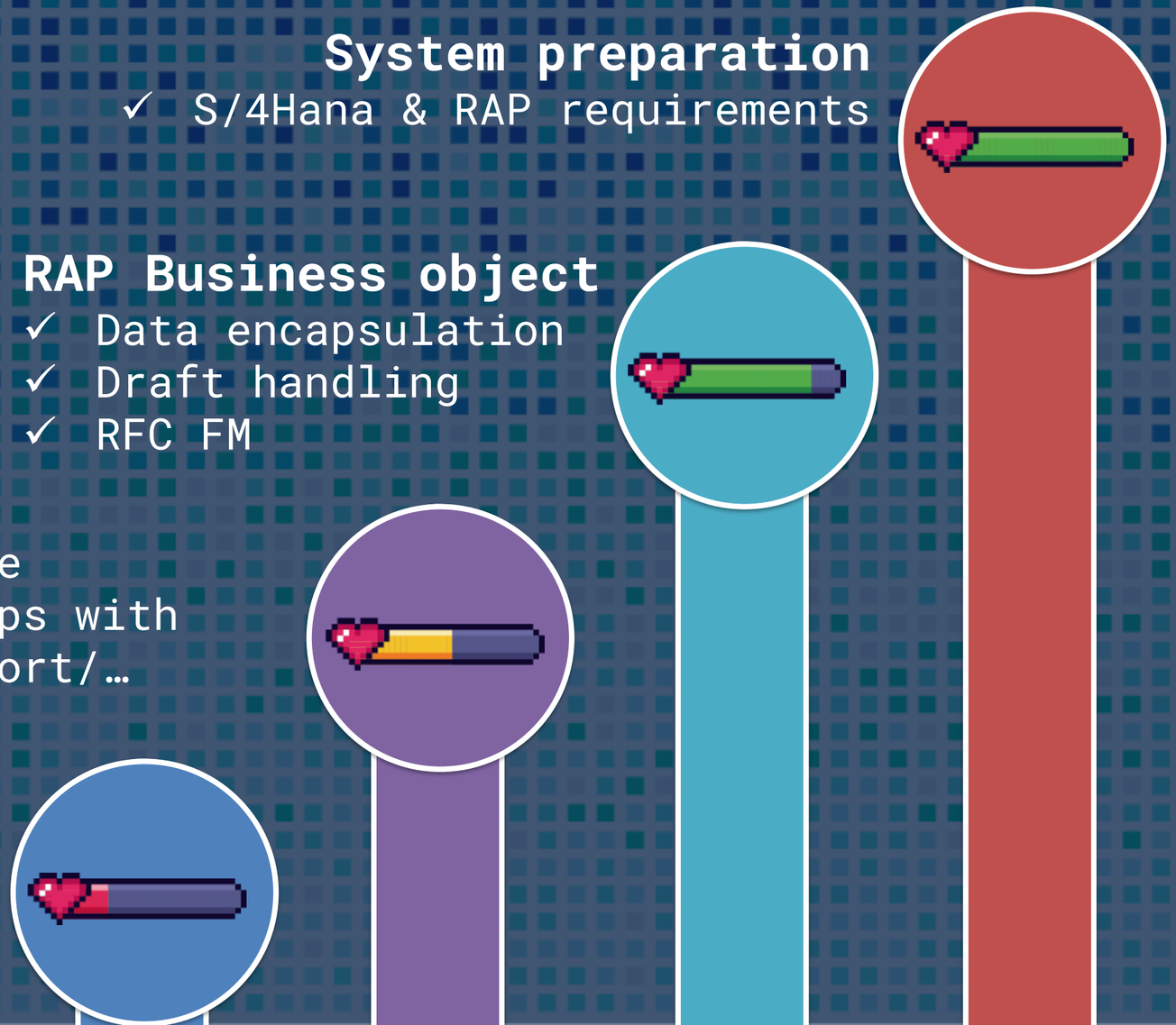
Conclusions

System preparation
- ✓ S/4Hana & RAP requirements

RAP Business object
- ✓ Data encapsulation
- ✓ Draft handling
- ✓ RFC FM

CDS
- ✓ Less code
- ✓ Valuehelps with filter/sort/…

Managed RAP
- ✓ CRUD out-of-the-box
- ✓ Enhanced messaging framework



How to get started….

CDS

Work with CDS (Core Data Services)

RAP + Web API

ABAP RESTful Application Programming Model

Just try it out!